

Evidence Selection in Budgeted Multi-hop Retrieval-Augmented Generation: From Proxy Objectives to Answer Quality

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Abstract

Evidence selection for multi-hop question answering must preserve complementary information within a limited reader context. This study examines the relationship between supervised evidence-set scores, the sets found by search, downstream answers, and computational cost. We compare first-, second-, and fourth-order factorization models with a general permutation-invariant set model, sharing frozen dense embeddings, paragraph candidates, annotation supervision, and a fixed reader. Strong accuracy on constructed set comparisons coexists with losses of complete annotated support during actual selection. A bounded continuation trains the deployed score on merged search frontiers and final sets, with an equal-update static replay comparator. For the primary seed, fourth-order support completeness rises from 0.4531 to 0.5156 on the checkpoint panel, but answer F1 falls from 0.4181 to 0.3766 on a separate 128-question development panel; maximal marginal relevance scores 0.4303. Neither seed yields an aligned model above MMR in F1 point estimate; lower-order responses differ from H4’s. We also separate search preservation from efficiency: granular-ball and k-means indexes retain flat-search choices but do not reduce measured latency at 128 candidates. Earlier sentence-level completion and fixed-evidence placement results are reported separately. The findings show how favorable proxy metrics can fail to establish practical value, without claiming independent confirmation or a general advantage of higher-order models, granular balls, or open-domain retrieval.

Keywords: Retrieval-augmented generation, Evidence selection, Multi-hop question answering, Set prediction, Search-aligned learning, Empirical evaluation

1 Introduction

A retrieval-augmented question-answering system must decide which evidence to place in its reader’s context. For multi-hop questions, individual relevance is only part of that decision: several passages may jointly supply a connection that no passage provides

alone. A large candidate list can contain the required facts and still yield an incomplete or distracting context after selection. Conversely, adding more retrieved material can consume a limited input budget without improving an answer.

This motivates evidence-set models, which score combinations rather than isolated passages. Recent retrieval work explicitly models collective information needs [1], and multi-hop beam retrieval retains multiple partial hypotheses [2]. Such designs raise a measurement question: does a model that recognizes good evidence sets also select them when optimized by its actual search procedure? The two tasks need not induce the same input distribution. Constructed positive and negative sets may be easy to distinguish, while search discovers sets that exploit inaccurate scores. The resulting selection must then pass a second test: helping a particular reader produce a correct answer.

We investigate these transitions in a controlled development study. Frozen BGE embeddings retrieve a shared list of 128 paragraphs. Four compact models receive the same known-support supervision: first-order (H1), second-order (H2), fourth-order (H4), and a general DeepSets control. All select under the same actual 1,024-token input limit, use the same six-block maximum, and feed the same unadapted FP16 reader. The analysis distinguishes sampled-set prediction, actual selected support, canonical answer quality, and measured cost. A bounded continuation then changes the training distribution and preference loss while retaining the model families, candidates, renderer, and search.

The investigation originates in HyperGranular-RAG, a sentence-level structured evidence-completion system. Its earlier compact-backbone gains are real but conditional: they did not establish an advantage over a stronger BGE retriever. The present paragraph-set models are a subsequent design, not a corrected implementation of that historical heuristic. Likewise, using granular balls to index conditional set scores changes search computation, not the selected evidence when the index is exact. Keeping these method identities separate prevents a later mechanism from being used to explain an earlier score retrospectively.

The article makes an empirical contribution. It documents a case in which strong support-proxy prediction fails to preserve support during search, tests a direct and bounded correction with simple and learned controls, and shows why exact indexing and reduced inner-product counts are insufficient evidence of acceleration. The evidence includes favorable, unfavorable, and unresolved comparisons. It does not establish that high interaction order is necessary, that granular balls are uniquely useful, or that one tested method generalizes to unrestricted open-domain RAG.

2 Related work and the scope of comparison

2.1 Retrieval, compression, and set selection

Dense passage retrieval represents queries and passages in a shared space for efficient relevance ranking [3]. Multi-hop methods such as MDR condition later retrieval on previously found evidence [4], while IRCoT alternates retrieval with generated reasoning [5]. Beam Retrieval jointly trains an encoder and hop-dependent heads to preserve multiple passage hypotheses [2]. Our candidate universe is fixed before set selection; we do not reproduce those systems or borrow their reported performance.

Compression and joint selection are also established research directions. RECOMP trains extractive and abstractive compressors for downstream language model utility [6]. SetR uses information-requirement reasoning and teacher-generated set selections [1]. Our models operate on cached embeddings and known support annotations, without an answer-based teacher or a generated summary. These differences define a low-cost experimental boundary. They do not establish superiority to compression or LLM-based set selection, which are not directly evaluated here.

MMR balances query relevance and redundancy [7]. Coverage-based summarization objectives similarly formalize complementarity [8]. Both are important alternatives to attributing a gain to higher-order structure. Our MMR baseline shares the candidate set, renderer, reader, and input limit with the learned selectors. A size-constrained version additionally supports like-for-like retrieval diagnostics.

2.2 Learned interactions and optimization of learned scores

Higher-order factorization machines provide low-rank ANOVA interactions and efficient recurrences for training them [9]. Deep Sets represents permutation-invariant functions through pooled member features [10]. We adapt these standard forms to query-conditioned evidence sets. Explicit fourth-order terms represent interactions among distinct members; they do not by themselves identify semantic relations or establish that a four-hop problem requires an order-four model.

The mismatch between training and search is not a new observation in machine learning. Beam-search optimization trains sequence scores on search violations [11]. Conservative objective models address the tendency of optimization to exploit overestimated regions of a learned objective [12]. Our intervention collects discrete evidence sets encountered by the existing selector and trains preferences from support annotations. It is neither a reproduction of sequence-level BSO nor the conservative objective in COMs, and inherits no guarantee from either. Their relevance is the need to evaluate scores where optimization will use them.

2.3 Granular and hypergraph representations

Granular-ball computing groups observations at different scales [13]. In retrieval, Cross-Granularity Hypergraph RAG uses entity–passage incidence and query-conditioned diffusion [14]; HyperGraphRAG retrieves extracted n-ary relations together with text [15]. These works precede general claims about combining structure with retrieval. We use the full name HyperGranular-RAG to avoid the shared HGRAG acronym, and do not equate its old lexical facet hyperedges with factual relations or learned ANOVA factors.

In the learned-selector study, a granular ball is an index node over query-conditioned scoring vectors. The maximum-inner-product bound is a standard center-plus-radius construction [16]. K-means receives the same vectors, bound, budget, and correctness requirements. Exact agreement with flat search establishes preservation, not an independent retrieval gain. Actual runtime determines whether the index is useful at the tested candidate count.

3 Task, evidence units, and information boundaries

3.1 Retrieval over the development paragraph corpus

HotpotQA and MuSiQue provide multi-hop questions with supporting evidence [17, 18]. We reuse 4,000 previously exposed questions and reconstruct whole paragraphs from their original sentence sources, retaining sentence-to-paragraph mappings. The resulting per-dataset corpora contain 9,791 HotpotQA and 31,130 MuSiQue blocks. A fixed BGE-large-en-v1.5 [19] encoder supplies normalized 1,024-dimensional query and paragraph vectors, and the top 128 paragraphs by dense score are shared by every selector. Evaluation never inserts a paragraph because it has a support annotation.

This corpus is broader than each question’s original closed candidate set, but it is not full Wikipedia. There is no ANN approximation in this candidate stage, and the corpora are assembled from the exposed benchmark materials. Source availability and the corpus construction therefore limit external validity. The BGE encoder uses its existing 512-token truncation rule: 16 HotpotQA and 8 MuSiQue blocks exceed that limit. The reader receives whole selected blocks under a separate input budget; encoder truncation and reader feasibility are different operations.

The group split contains 3,166 FIT questions, 738 TUNE questions, and 96 historical-development questions. FIT has 3,165 groups. Question groups are not split across roles, but the roles are not document-disjoint: FIT and TUNE share 57 HotpotQA and 4,753 MuSiQue source documents. All are exposed development material. In particular, excluding the answer-evaluation panel from checkpoint selection does not retrospectively turn it into an independent test set.

3.2 Feasibility and support labels

For question q , let V_q be the dense candidate list and $S \subseteq V_q$ a set of paragraph indices. Blocks are rendered with their titles and original text, in BGE order. Define

$$\mathcal{F}_q = \{S \subseteq V_q : |S| \leq 6, T(q, S) \leq 1024\}, \quad (1)$$

where T is the actual tokenizer length of the complete reader input, including chat framing, question, and protocol instructions. We neither truncate individual selected paragraphs nor assume that summing separately tokenized lengths exactly predicts T . The cardinality and token limits are both enforced. A long block may be infeasible even when a cardinality slot remains available.

Let G_q be the set of known support annotations, and let $C(S)$ be the annotations recovered through the original source mapping. We define

$$\text{cov}(S) = \frac{|C(S) \cap G_q|}{|G_q|}, \quad \text{full}(S) = \mathbf{1}[G_q \subseteq C(S)]. \quad (2)$$

Full support is stricter than mean coverage. Neither is a complete label of semantic usefulness: an unannotated paragraph can provide an alternative derivation, and a reader can fail despite seeing all annotated support. Targets supervise FIT sets and

evaluate outputs; they are not inputs to the deployment scorer, candidate ranking, or token feasibility check.

On all 738 TUNE questions, Top128 contains all known support for 190/196 HotpotQA and 367/542 MuSiQue questions. Known support fits the six-block and input-token constraints for all 196 HotpotQA questions and 533/542 MuSiQue questions, considered from their source blocks. These are separate diagnostics: support may fit yet be absent from Top128. We retain the nine budget-infeasible MuSiQue questions and every unreachable query in unconditional results.

4 Scoring sets and searching under the reader budget

4.1 Common representation and model families

Let $e_q, e_i \in \mathbb{R}^{1024}$ be normalized embeddings and l_i the standalone paragraph token length. Each query–paragraph feature vector concatenates $e_q, e_i, e_q \odot e_i, |e_q - e_i|$, dense cosine similarity, and $\log(1 + l_i)$. The last two features use FIT normalization statistics. A linear layer, GELU, and layer normalization produce $h_i \in \mathbb{R}^{128}$; a query network produces $h_q \in \mathbb{R}^{128}$. No answer, query ID, source support identity, or rank label enters these features.

Unary outputs a_{it} and query-conditioned cardinality biases $b_{qt}(k)$ are shared by the two prediction heads t : full support and coverage. For interaction order H , factors $v_i \in \mathbb{R}^{32}$ use a linear projection, layer normalization, and tanh. Query weights $w_{qth} \in \mathbb{R}^{32}$ score order h . The set logit is

$$g_{Ht}(S) = b_{qt}(|S|) + \frac{1}{6} \sum_{i \in S} a_{it} + \sum_{h=2}^H \frac{w_{qth}^\top e_h(S)}{32 \binom{6}{h}}, \quad (3)$$

where

$$e_h(S) = \sum_{i_1 < \dots < i_h, i_j \in S} v_{i_1} \odot \dots \odot v_{i_h}. \quad (4)$$

H1 has no interaction sum, H2 includes pairs, and H4 includes orders two through four. The denominator is fixed by the maximum cardinality six, not recomputed from $|S|$. An auxiliary coverage prediction is a separate head; only the full-support logit ranks deployed sets.

The general-set control retains unary and cardinality outputs and adds $\rho([\sum_{i \in S} h_i/6, h_q])$, a two-layer MLP with hidden width 192 and two outputs. This is a DeepSets-style architecture, not a reproduction of an LLM-based set selector. Parameter counts are comparable but not identical: H2 has 670,640, H4 687,152, and DeepSets 707,922 parameters. H1 registers 662,384 parameters, of which 4,192 interaction-projection parameters do not affect its score. Its effective scored parameter count is 658,192. We do not conceal that inactive branch as useful first-order capacity.

4.2 Static training and its evaluation target

The initial training collection includes complete known-support sets, sets with one known support removed, same-size replacements, redundant additions, Dense-K6/MMR-K6, and bounded random examples. FIT-only reference construction may append missing support blocks and matched distractors to form supervised sets. That injection is limited to static training examples. Actual evaluation and search mining always use the original Top128 candidates.

The base objective combines binary cross-entropy for full support, 0.5 times coverage MSE after a sigmoid, 0.25 times pointwise support classification, and 0.25 times a softplus ranking term for constructed same-cardinality complete/incomplete pairs. Pointwise support labels again mean annotation membership, not factual correctness. Initial checkpoints were selected using original TUNE loss. Three initial training seeds (1729, 2026, and 426) are retained. They share the same questions, inputs, and answer-evaluation panel.

The constructed pair task is easier to control than deployment. It fixes the alternatives presented to the model, often around known supports. Search instead asks the model to choose among many feasible combinations. Consequently, a high pair accuracy can be compatible with a low rate of complete support in the selected output. We measure both endpoints rather than treating the first as a substitute for the second.

4.3 Actual Flat search and simple selectors

The common learned search starts from the empty set. At each of six depths, each surviving parent proposes its four highest-scoring feasible distinct extensions, and the global beam retains four unique sets. All visited sets, including those not retained for further expansion, remain eligible for final selection. A feasible Dense-K6 set is also included. The final choice maximizes the deployed full-support logit, with fewer actual input tokens and then stable block identities breaking ties. Search does not terminate merely because the next score increment is negative. This procedure is exact with respect to each tested extension ranking, but the beam does not enumerate all feasible sets and is not globally optimal.

Dense-budget greedily considers paragraphs in their initial dense order, accepting a block when the complete input still fits. MMR-budget repeatedly chooses the largest $0.7 e_q^\top e_i - 0.3 \max_{j \in S} e_i^\top e_j$ among remaining candidates, accepting it if feasible. Both skip overlong additions and can retain more than six short blocks. Their K6 versions impose the same cardinality as learned search. All selected contexts are finally rendered in original BGE order, so MMR’s selection sequence does not grant it a different placement policy.

4.4 What exact ball search can guarantee

The ANOVA recurrence updates e_h by descending order, $e_h \leftarrow e_h + v_i \odot e_{h-1}$, with $e_0 = \mathbf{1}$. This is the standard higher-order factorization computation [9]. For fixed parent S , the score of adding i can be written

$$g_H(S \cup \{i\}) = o_q(S) + \theta_q(S)^\top z_i, \quad (5)$$

where $z_i = [a_i/6, v_i]$ and the offset includes the parent score and cardinality-bias change. Thus the conditional extension ranking is a maximum inner-product problem in a learned query-specific space.

For a group with center c_B and maximum Euclidean radius $r_B = \max_{i \in B} \|z_i - c_B\|_2$, Cauchy–Schwarz gives

$$\max_{i \in B} \theta^\top z_i \leq \theta^\top c_B + \|\theta\|_2 r_B. \quad (6)$$

The proof is immediate from $z_i = c_B + (z_i - c_B)$ and the definition of r_B . An average cosine radius would not justify this bound. The granular-ball index recursively partitions the scoring vectors with farthest pivots; the k-means control uses the same space and bound with iterated two-center assignments. Both preserve every original member and perform the same token feasibility checks. They prune only groups unable to improve the current feasible extension threshold, retaining ties and numerical boundary cases.

Equation 6 is a standard search bound [16], not an answer-quality theorem. Even exact preservation offers no rescue when the score chooses an incomplete set. Nor does pruning prove a speedup: factor construction, per-query indexing, traversal, and tokenizer work all contribute to latency. These costs are measured separately in Section 8.2.

5 A bounded search-aligned correction

5.1 Mining the shared error distribution

We select 512 FIT questions by deterministic group hash, with 256 per dataset and proportional allocation to existing question types. Every architecture searches these questions using its current score. The mined pool merges final sets, high-scoring partial frontiers, Dense-K6/MMR-K6, and at most eight deterministic feasible replacement attempts per model trajectory. The merged, deduplicated pool is capped at 64 sets per question. Model outputs are labeled from actual source identities, and are not presumed incorrect merely because they were selected by a model.

All four architectures receive the same merged pool. No method receives exclusive teacher calls, stronger candidates, or a different reader. Unreachable queries still supply differences in partial annotated coverage; we neither delete them nor invent a complete positive set. A possible second round re-mines the current selected models rather than merely replaying the first pool. The entire intervention is capped at two rounds.

The primary-seed pools retain 64 sets for each of the 512 queries in each round. A missing preference is not fabricated: round one has no eligible final-set pair for 171/256 HotpotQA and 115/256 MuSiQue queries, and no frontier pair for 17/256 and 42/256. In round two these counts are 196/256, 135/256, 19/256, and 59/256, respectively. Pool capacity prioritizes model finals and high-scoring frontiers, so attempted replacements need not survive the cap. This limits the final-pair supervision actually available, even though every query continues to contribute base observations.

5.2 Preferences that train the deployed score

Define the training preference $u(S) = 0.5 \text{ full}(S) + 0.5 \text{ cov}(S)$. For $u(S^+) > u(S^-)$, use

$$\ell = \log(1 + \exp\{0.2[u(S^+) - u(S^-)] - [g(S^+) - g(S^-)]\}) . \quad (7)$$

Final-set comparisons prefer equal cardinality and token-length ratios in $[0.8, 1.25]$, with same-cardinality fallback when no near-length pairs exist. Frontier comparisons prefer the same parent and depth; same-depth comparisons are a fallback when those pairs provide no preference. Missing pair categories are masked and counted. The deployed logit g , rather than only the auxiliary coverage head, receives gradients even when both partial sets have zero full-support labels.

The total loss is

$$L = L_{\text{base}} + 0.5L_{\text{final}} + 0.5L_{\text{frontier}} . \quad (8)$$

Each query supplies four static and four mined base observations and up to one sampled pair from each added category. Base and preference terms are normalized within query and then across queries. More enumerated states therefore do not automatically increase a query’s training weight. The preference is deliberately an annotation utility, not a reader reward. No generated answer F1 is used to form pairs or choose a checkpoint.

5.3 Controls, checkpoints, and stopping

Continuation starts from the corresponding original checkpoint with AdamW, learning rate 3×10^{-4} , weight decay 10^{-4} , query batch eight, and gradient clipping at one. Each round permits eight epochs over the same 512-query mined FIT subset. H4 static replay uses eight static sets per query, the original loss, and the same optimizer-update budget. Its reported checkpoint matches the selected aligned H4 checkpoint’s update count, including the corresponding preceding-round starting point when applicable. It is not independently chosen for favorable QA performance.

A disjoint-from-QA checkpoint panel contains 128 TUNE questions, 64 per dataset. At epochs 2, 4, and 8 we run actual selection. The fixed lexicographic criterion maximizes dataset-equal full support, then coverage, then minimizes original development loss, with earlier checkpoints breaking remaining ties. The fixed natural-QA panel is excluded from this decision. Because all materials have been exposed during development, the study is still exploratory.

Both seeds (1729 and 2026) completed two rounds. The second seed was triggered by selection-panel improvement before any new QA score was observed; all four architectures received the same continuation.

The protocol caps new GPU process wall time at two hours, measured CPU process time at three hours, and new reader calls at 1,200, subject to the remaining cumulative budget. At least one quarter of the GPU allowance is reserved for primary QA. Optional second-seed work applies to all four aligned models only when actual checkpoint-panel selection improves and resources permit. Stop rules prohibit another model family, a larger beam, an answer reward, a new data boundary, or a later experiment round in order to obtain a preferred outcome. The achieved scope and costs are reported with the results.

6 Evaluation and reproducibility

6.1 Reader and canonical scores

The reader is the original Qwen2.5-3B-Instruct checkpoint [20] in FP16, without the relation-front-end adapter. Its P2 interface requests a single answer field, uses constrained decoding and greedy generation, and permits 128 new tokens. Every method shares the tokenizer, chat framing, original-text renderer, precision, and input limit. A public protocol improvement is not attributed to granular balls or higher-order scoring.

The main panel has 128 questions: 48 HotpotQA bridge and 16 comparison questions, and 32 MuSiQue two-hop and 32 three/four-hop questions. Canonical dataset-specific EM and token F1 score the parsed answer field, including each dataset’s normalization and alias conventions. We do not apply HotpotQA-specific yes/no handling indiscriminately to MuSiQue, and do not clean answers using the target. Invalid output remains a failure in the full denominator. UNKNOWN, EOS, output caps, and format failures are recorded separately.

The primary continuation comparison retains Dense, MMR, the original H4, static H4 replay, and H1/H2/H4/DeepSets aligned models. A logical result can reuse a prior generation only when full input token IDs, model, precision, generation configuration, and constrained-decoding identity match. Merely sharing evidence text does not satisfy that condition. Cache reuse is reported as reuse, not independent inference or a deployment latency advantage.

6.2 Interpretation of repeated comparisons

Dataset means are reported separately and combined with equal dataset weight. The same questions appear across methods and training seeds. Seed standard deviations describe training variation, not confidence intervals over independent questions. The continuation does not introduce new confirmation p-values, and no nonsignificant or close comparison is called equivalent. We retain failures, unreachable support, and contexts that cannot accommodate all reference evidence. Conditional diagnostics are labeled and do not replace unconditional means.

Historical sentence-completion results in Section 9 use their existing canonical rescoring and original paired-bootstrap intervals. That additive re-evaluation fixes predictions rather than generating new ones and is not a new independent confirmation. Those intervals are not transferred to the current learned-selector comparisons.

7 Results: from proxy fit to actual selection

7.1 The original mismatch

On constructed same-size complete/incomplete pairs, H4 achieves accuracy 0.9645 on HotpotQA and 0.9464 on MuSiQue. Yet its actual selected support completeness on all 738 TUNE queries is 0.7092 and 0.2232, below Dense-K6’s 0.8214 and 0.2565. Table 1 includes the lower-order and general set alternatives. H4’s favorable pair metric therefore cannot establish a practical selection advantage.

Table 1 Original static models (seed 1729) on the 738-query TUNE boundary. Pair accuracy uses constructed same-cardinality complete/incomplete pairs. Full support uses the actual chosen set. H/M denote HotpotQA/MuSiQue.

Method	Pair H	Pair M	Full H	Full M
Dense-K6	—	—	.8214	.2565
H1	.9475	.8978	.6531	.2454
H2	.9638	.9422	.6429	.2325
DeepSets	.9620	.9482	.6990	.2306
H4	.9645	.9464	.7092	.2232

Table 2 Checkpoint-panel support completeness before and after search alignment (64 questions per dataset). Selection uses this panel, not QA.

Seed	Model	Static H	Aligned H	Static M	Aligned M
1729	H1	0.5938	0.7500	0.2656	0.2656
1729	H2	0.6406	0.7500	0.2344	0.2500
1729	DeepSets	0.6719	0.7500	0.2188	0.2656
1729	H4	0.6719	0.7500	0.2344	0.2812
2026	H1	0.6094	0.7656	0.2344	0.2656
2026	H2	0.6406	0.7188	0.2500	0.2500
2026	DeepSets	0.6250	0.7812	0.2344	0.2656
2026	H4	0.6719	0.7188	0.2344	0.2500

The loss is visible at the query level. Relative to Dense-K6, H4 loses full support on 27 HotpotQA and 42 MuSiQue queries while recovering it on only 5 and 24. The selected set scores are at least as high as the included Dense-K6 scores, so this is not caused by omitting that fallback from the score comparison. Mean sigmoid outputs on selected sets are 0.9920 and 0.9667, far above their observed full-support frequencies. We describe these as model outputs, not calibrated probabilities. The scorer is optimizing its learned criterion successfully while that criterion can favor poorer annotated evidence.

Additional controls limit the interpretation of high pair accuracy. On exact-length and same-cardinality matched comparisons, H4 accuracy is 0.8333/0.7878, compared with DeepSets 0.9524/0.8525. These are smaller diagnostic subsets: 65 HotpotQA queries with 84 pairs and 185 MuSiQue queries with 278 pairs. They do not replace the full selection boundary, but weaken an explanation based on uniquely superior higher-order discrimination. High-order parameters receive nonzero gradients and contribute nonzero terms, so their existence in the model is not merely nominal. Their activation still does not establish their necessity.

Table 3 Canonical answer F1 on the fixed 128-query development panel (64 per dataset), at 1,024 input tokens. Original rows reuse their exact prior generations.

Seed	Method	HotpotQA	MuSiQue	Mean
1729	Dense	0.6324	0.1786	0.4055
1729	MMR	0.6615	0.1991	0.4303
1729	Static H4	0.6341	0.2021	0.4181
1729	H4 replay	0.6492	0.1600	0.4046
1729	H1 aligned	0.6415	0.1714	0.4065
1729	H2 aligned	0.6188	0.1927	0.4057
1729	DeepSets aligned	0.6136	0.2044	0.4090
1729	H4 aligned	0.5902	0.1630	0.3766
2026	H1 aligned	0.6289	0.1962	0.4126
2026	H2 aligned	0.6482	0.1865	0.4173
2026	DeepSets aligned	0.5777	0.1266	0.3521
2026	H4 aligned	0.6084	0.1578	0.3831

7.2 Search-aligned development results

Table 2 shows actual checkpoint-panel selections. For seed 1729, H1 changes equal-dataset completeness by +0.0781, H2 changes equal-dataset completeness by +0.0625, DeepSets changes equal-dataset completeness by +0.0625, H4 changes equal-dataset completeness by +0.0625. These are selected development effects, with the full checkpoint trajectory retained. They do not by themselves establish answer improvement or a unique advantage of fourth-order interactions. Every reported aligned checkpoint comes from round one; the completed second rounds did not improve the best selection criterion. Continued loss reduction was not continued selection progress.

8 Results: natural answers and computational cost

8.1 The fixed natural-QA panel

Table 3 reports answer F1. For primary seed 1729, aligned H4 changes mean F1 from 0.4181 to 0.3766. Its difference from MMR is -0.0537 and from matched static replay is -0.0281 . For seed 2026, H4 changes from 0.4304 to 0.3831; aligned H1, H2 and DeepSets score 0.4126, 0.4173 and 0.3521. Every method retains all 128 exposed questions. These descriptive comparisons do not establish independent confirmation. Relative to each corresponding static checkpoint, the primary-seed mean F1 changes are H1 +0.0326, H2 +0.0060, DeepSets -0.0033 , H4 -0.0415 . For seed 2026, the corresponding changes are H1 -0.0151 , H2 +0.0051, DeepSets -0.0505 , H4 -0.0473 . The static comparisons reuse the same original questions and predictions; no additional generation or independent replication is implied.

Table 4 separates the cardinality actually used, the full tokenized input, annotated support, and exact answer match. The nominal 1,024-token cap is shared, not a claim

Table 4 Primary-seed QA-panel selection and output diagnostics, equal dataset weight over all 128 questions. Full is annotated support completeness; EM is canonical answer exact match. Input includes instructions.

Method	Full	Coverage	Blocks	Input tokens	EM
Dense	0.5000	0.7096	6.95	1009.9	0.3359
MMR	0.4766	0.7025	7.25	1007.3	0.3672
Static H4	0.4297	0.6608	5.91	916.9	0.3594
H4 replay	0.4297	0.6647	5.91	900.9	0.3359
H1 aligned	0.4766	0.6960	4.45	696.2	0.3359
H2 aligned	0.4375	0.6667	5.53	789.7	0.3516
DeepSets aligned	0.4688	0.6914	4.43	693.4	0.3516
H4 aligned	0.4453	0.6803	5.70	842.2	0.2969

of identical input lengths. Dense and MMR may use more short paragraphs than the learned six-block selectors.

The original three-seed results provide context without being pooled with the continuation as new replications. Dataset-equal mean F1 is 0.4048 for H1, 0.4109 for H2, 0.4041 for DeepSets, and 0.4171 for H4. Dense and MMR score 0.4055 and 0.4303. H4’s seed range is 0.4027–0.4304, and H1/H2 exceed it for seed 426. Thus a favorable single checkpoint does not establish a general interaction-order advantage. The fixed reader’s response can change when an annotation-equivalent set contains different context, so support and answer comparisons must both remain visible.

Earlier 64-query token-budget sensitivity also remains a separate boundary. At 512/1,024/2,048 tokens, H4’s equal-dataset F1 is 0.4020/0.4649/0.4857, while MMR’s is 0.4281/0.4489/0.3737. These measurements use the same smaller subset across token budgets, not the full 128-question panel. They suggest dependence on context and selection policy but do not license choosing 2,048 tokens post hoc for the present main result. The continuation retains the prescribed 1,024-token budget and does not run a new budget grid.

8.2 Search correctness and latency

Exact granular-ball and k-means search agree with Flat on 1,476 comparisons over all 738 TUNE queries and 30,956 tested parent states. A bounded implementation improvement that skips demonstrably inferior exact leaf scores was subsequently tested on a fixed 32+32-query subset, producing 128 comparisons and 2,688 parent states with no mismatch. Table 5 reports that latter timing boundary. The initial full-boundary timings are also retained in the reproducibility materials rather than overwritten.

Granular balls reduce inner-product evaluations by approximately 66% on HotpotQA and 44% on MuSiQue, but their measured latency remains 27.2% and 19.5% above Flat. Both tree types incur construction and traversal overhead and can require more token checks before establishing feasible incumbents. All candidate embeddings and small-network factors are still computed. Some CPU measurements overlap GPU

Table 5 Existing fixed-subset mean selection time in seconds per query, including network, per-query index construction, and tokenizer feasibility. These component timings precede the search-aligned training intervention.

Dataset (queries)	Flat	Granular ball	K-means
HotpotQA (32)	.33459	.42557	.42268
MuSiQue (32)	.54612	.65282	.64392

reader work, which precludes claims of isolated device-independent speed. The evidence supports exact preservation at this boundary, not acceleration.

Replacing each leaf by one permanent representative is a different, approximate method. It changes all 738 selected sets and reduces complete support to 0.3469/0.0535. This diagnostic explains why preserving every raw member matters here. It does not refute every approximate grouping method, and no additional reader outputs were generated for that control.

8.3 Actual experiment expenditure

The continuation records 4,513.07 GPU process-wall seconds, 6,997.06 measured research CPU seconds, and 986 real reader calls, including eight uncached checks. It produces 1,536 logical QA rows with 558 cache hits. New generations total 761,847 input tokens and 9,460 output tokens. No comparison row has an invalid answer field or reaches the output limit. Typesetting CPU is accounted separately in the accompanying ledger. Older unmeasured CPU use remains unknown; phase accounting is not reconstructed whole-system billing. No paid compute is used. The prior static-selector study used 4,350.28 GPU process-wall seconds, 10,541.75 measured CPU seconds, and 2,287 real reader calls including eight uncached checks. Its 3,520 logical rows contain 1,241 cache hits. Those rows are neither 3,520 independent examples nor 3,520 fresh generations. Costs for old relation-search prototypes do not disappear when their outputs are reused, but they are not charged as new calls in the present continuation.

9 Context from earlier evidence-completion studies

The earlier static-q25 system used sentence units, compact MiniLM geometry, query-term facet relations, and at most four inserted units after ten protected dense entries. It retained a final Top-20 entry budget under a separate 4,096-token input cap and used Qwen2.5-1.5B-Instruct. Its facet proposal was scored once relative to seed groups; it was not the dynamically recomputed learned set score of this article. Table 6 preserves canonical effects and recorded intervals under their original boundaries.

Three compact-backbone answer improvements are supported within those closed-candidate conditions. The specified facet selector also exceeds its centroid-only control, but that comparison does not isolate every possible coverage or diversity heuristic. The original compact Full system is worse than BGE. Neither its cross-space sidecar nor a BGE-native reconstruction establishes an increment over BGE, and the Gemma

Table 6 Historical sentence-completion evidence, kept separate from the paragraph-selector experiments. Effects are answer-F1 differences, with recorded 95% paired-bootstrap intervals. Joint studies use equal dataset weight. Unmarked component/BGE/placement rows contain 2,500 queries on their respective boundaries; Gemma transfer contains 4,000. Each contrast retains its original full denominator; different boundaries are not pooled.

Comparison	$\Delta F1$	95% interval
Compact minus Dense, HotpotQA (1,000)	.01478	[.00020,.02988]
Compact minus Dense, MuSiQue (3,000)	.01140	[.00450,.01835]
Compact minus Dense, joint (2,500)	.01357	[.00491,.02233]
Compact Full minus NoFacet	.01336	[.00341,.02343]
Original compact Full minus BGE	-.03998	[-.05393,-.02621]
BGE sidecar Protected minus BGE	-.00256	[-.00998,.00458]
BGE-native Protected minus BGE	-.00305	[-.00688,.00063]
Gemma transfer, completion minus Dense	.00516	[-.00262,.01295]
Cross-space Protected minus Unprotected	.01122	[.00129,.02104]

transfer interval crosses zero. These unresolved effects are not evidence of equivalence or of universal ineffectiveness.

The original Full versus NoProtection ablation remains unresolved ($\Delta F1 = 0.00354$, recorded interval $[-0.00675, 0.01389]$). It does not establish the independent necessity of the old protection component. This ablation and the later fixed-content ordering comparison test different interventions and must not be substituted for one another.

The cross-space placement comparison holds prompt-visible evidence content fixed and changes its ordering. With outside-Top20 insertions I , $|I| = m \leq 4$, both policies contain $I \cup \{d_1, \dots, d_{20-m}\}$, and the completed position audit verified the visible text and truncation conditions. Its positive effect therefore describes the tested ordering/placement policy on all original queries, not an internal attention mechanism or a favorable post hoc subset. A policy can improve how a reader uses a fixed set without establishing that the set improves BGE’s own evidence.

These results motivate studying evidence selection more directly, but they cannot validate the new learned models. Conversely, a successful search correction would not retroactively prove that the old granular-ball or facet construction caused the compact gains. The two studies differ in candidate universe, unit granularity, generator, prompt, and budgets.

10 Discussion

10.1 A score must be evaluated under its optimizer

The constructed-set results show that the models learn a useful supervised distinction, but the deployment results identify a different requirement. Optimization changes the distribution of evaluated sets. A model may recognize a supplied missing-support contrast while assigning excessive scores to other combinations that its search discovers. Including Dense-K6 as a feasible candidate makes the failure particularly informative:

the search can reject an available, more complete set because the deployed criterion ranks it lower.

The continuation targets this discrepancy directly. Its partial-state preferences can change the actual search order even when no set in a comparison is fully complete. The static replay control distinguishes that change from additional optimizer steps. Low-order and general-set controls test whether explicit fourth-order interactions are responsible for any gain. An improvement shared by those controls is evidence for training alignment within this setup, not evidence for a unique granular or hypergraph mechanism.

10.2 Evidence quality and reader quality remain distinct

Support annotation offers inexpensive supervision with interpretable labels, but it is an incomplete account of useful context. Alternative evidence, entity disambiguation, distracting text, and the reader’s own knowledge can all change an answer without changing known-support completeness. Conversely, complete annotated support does not guarantee that the reader will combine the facts correctly. Reader protocol and constrained-output changes belong to the public interface and must not be attributed to the evidence model.

This distinction also limits a negative interpretation. A model’s failure to improve F1 on one exposed panel does not prove that its evidence is useless on all readers. Equally, a positive point estimate cannot establish generalization after repeated development. The appropriate conclusion is the effect actually measured under the shared candidate, budget, and reader conditions, together with the controls that explain or fail to explain it.

10.3 Indexing is a systems hypothesis

Exact indexing asks whether the same answer to a search problem can be obtained more cheaply. It does not add evidence-selection capacity when it preserves the scorer and beam. At only 128 candidates, vectorized Flat search has small arithmetic cost, while index traversal and actual-token feasibility remain visible. The ball and k-means results therefore favor Flat in this implementation. A larger corpus or another feasibility regime might change the balance, but no such result is claimed here.

The practical lesson is to evaluate a sequence of interfaces rather than one attractive metric: supervision must train the deployed score; the score must choose useful evidence; the reader must use it; and any acceleration must include the costs it introduces. This decomposition is applicable as an evaluation strategy beyond the present model family, although its empirical outcomes are bounded to the tested system.

11 Limitations

All learned-selector and continuation results use exposed development materials. There is no new independent confirmation, full-Wikipedia run, new language, or external-method deployment. Document overlap between FIT and TUNE prevents unseen-document claims. The answer panel is small and balanced by design rather than

representative of an arbitrary deployment distribution. Training-seed variation cannot replace query-level external validation, and resource-bounded continuation does not rule out all future learning objectives.

Known supports are incomplete semantic labels. Some required blocks are unreachable or too long, and the encoder itself truncates a small number of paragraphs. We report these boundaries rather than labeling every lost or unannotated passage as noise. The use of a single 3B reader and a fixed structured-output protocol limits generator conclusions. Earlier generator transfer remains unresolved under its own conditions.

The study does not reproduce SetR, RECOMP, Beam Retrieval, or a large graph-RAG system under matched conditions. It supplies simple and internal learned controls, not a comprehensive leaderboard. The historical facet comparison leaves ordinary clustering and direct selection attribution unresolved. Standard ANOVA algebra and maximum-radius bounds are included to define the evaluated algorithms; they are not new theoretical contributions or guarantees about answer F1.

12 Conclusion

Budgeted evidence selection should be judged where its score is used. In this study, strong constructed-set prediction coexists with losses of annotated support during search, and higher-order modeling does not by itself establish an answer-quality advantage over MMR. A bounded continuation improves checkpoint-panel support selection without changing the reader or candidate universe, but H4 answer F1 declines in both training seeds and every aligned architecture remains below MMR in F1 point estimate. H2 improves slightly over its own static checkpoint in both seeds, while H1 improves in only the primary seed. The answer effects therefore depend on architecture and seed. Exact granular-ball and k-means indexes preserve selection but provide no measured acceleration at 128 candidates. Together with the separately reported historical completion and placement results, the evidence supports an empirical account of the gap between proxy objectives, selected context, answers, and cost. Stronger deployment claims require new prospective evidence, not reinterpretation of the development comparisons reported here.

Statements and declarations

Data and code availability.

The study uses the cited public benchmark sources and existing local model snapshots. Reproducible code and public aggregate tables accompany the manuscript. Source text, answer annotations, model weights, and caches are not bundled in the public summaries. Exact local artifacts retain the question, source, selection, checkpoint, and generation identities required for tracing reported values. A final persistent archival link and redistribution terms require author confirmation.

AI assistance.

An AI assistant contributed implementation, analysis, targeted literature checking, and manuscript drafting and revision. This extends beyond copy editing. Human authors

must verify the final evidence and text, approve submission, and take responsibility for the work. The manuscript’s mentor-style and reviewer-style checks are two perspectives of the same assistant, not independent expert peer review.

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